

ASYN4 Machine Performance Report

Field	Value
NAME	CAGE MOTOR
REKE	SC317
DATUM	17.09.26

Partial-Load Performance

Load	P_mech (kW)	Eta	cos(phi)	I1 (A)	Slip (%)	Pcu1 (kW)	Pcu2 (kW)	M (Nm)
1/4	90.05	0.9240	0.824	94.51		1.718	3.070	1480.1
2/4	22.50	0.8828	0.420	48.44		0.451	0.202	360.8
3/4	44.99	0.9216	0.650	60.08		0.694	0.745	727.2
4/4 (rated)	67.48	0.9275	0.766	75.92		1.109	1.682	1099.5
5/4	112.87	0.9159	0.853	115.38		2.560	4.999	1873.2

Breakdown / Locked-Rotor Summary

Quantity	Value
Breakdown slip (Kipp)	21.887 %
Breakdown torque	4811.4 Nm
Locked-rotor torque (theoretical)	3327.9 Nm
Locked-rotor current (theoretical)	554.1 A
Locked-rotor torque (corrected, KKA=0.50)	4331.1 Nm
Locked-rotor current (corrected)	634.9 A
Time constant TK	0.0242 s
Time constant THAAPT (rotor)	0.1872 s
Time constant TSTREU (leakage)	0.0073 s

Cooling & Material Information

Quantity	Value
Dominant cooling path (IKLINK)	0
Stator yoke iron area AR1	0.04243 m2
Stator tooth iron area AZ1	0.02236 m2
Rotor yoke iron area AR2	0.09214 m2
Rotor tooth iron area AZ2	0.01844 m2